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Paleomagnetism of Miocene sediments from the Turfan Basin, Northwest China: no significant vertical-axis rotation during Neotectonic compression within the Tian Shan Range, Central Asia

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Abstract

Paleomagnetic results are reported from 377 samples of Miocene red sediments in the Turfan Basin of East Tian Shan, Northwest China. Thermal demagnetization for most specimens isolates a low-temperature component that groups around the present geomagnetic field. A high-temperature characteristic component is isolated from four localities and interpreted as primary remanent magnetization from apparent pre-folding origin and positive reversal test. As for previous results obtained from Tertiary rocks in central Asia, the magnetic inclination is significantly shallower than the reference field computed from the Eurasian APW path at 10 and 20 Ma. Magnetic susceptibility measured in 324 specimens exhibits low anisotropy of less than 1.05 for >80% of the sample. The low anisotropy degrees, AMS shape and orientations suggest that the studied rocks are unlikely to have undergone significant strain during compaction or tectonism. We argue that the shallow inclination is likely to result from an initially shallow detrital remanent magnetization. Comparison of the paleomagnetic declination from Eurasia, North China, and Tarim suggests that the Turfan Basin has been subjected to little or no vertical-axis rotation since the Miocene. Together with paleomagnetic declinations on both sides of the Talas-Ferghana fault, we propose that the Tian Shan to the east of the Talas-Ferghana fault has remained relatively stationary with respect to adjacent blocks since the Miocene.
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Keywords: Turfan Basin; Tian Shan; Miocene; Paleomagnetism; Inclination shallowing; Vertical-axis rotation

1. Introduction

Western China is interpreted as an assemblage of blocks, namely Lhasa, Qiangtang, Kunlun, Qaidam, Tarim and Junggar (Fig. 1a), which has undergone a complicated tectonics related to continental collision and intracontinental deformation since the Carboniferous (e.g. Molnar and Tapponnier, 1975; Tapponnier and Molnar, 1979; Besse et al., 1984; Patriat and Achache, 1984; Lin and Watts, 1988; Avouac et al., 1993; Hendrix et al., 1992; Chen et

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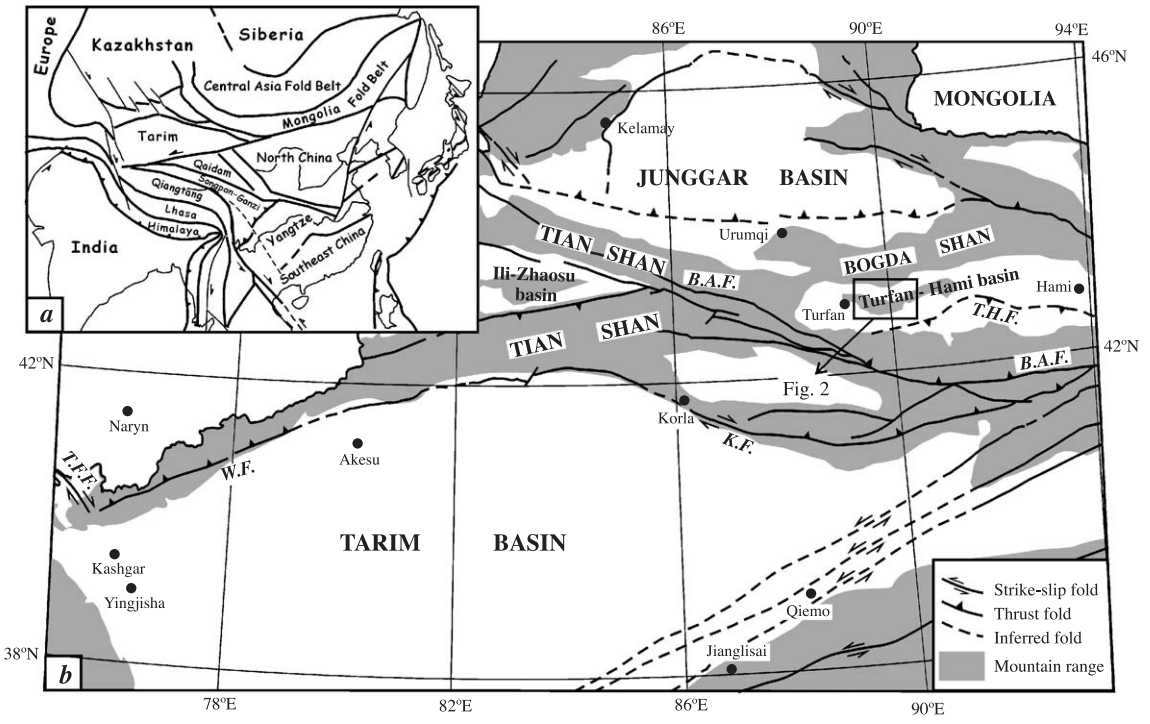


Fig. 1. Schematic maps of (a) the present location of the continental blocks of China showing the tectonic framework and (b) the Turfan Basin and surroundings after the tectonic map of Xinjiang, scale 1:2000000 (BGMRX, 1993). Abbreviations are: B.A.F., Bolokenu-Aqikekuduke fault; K.F., Korla fault; T.F.F., Talas-Ferghana fault; T.H.F., Turfan-Hami fault; W.F., Wuqia fault.

al., 1993; Gilder et al., 1996). Studies over three decades show that lithosphere thickening and intracontinental shortening are prominent in fold belts between these blocks or microplates and have resulted in formation of the present Tibetan plateau and hinterland (e.g. Molnar and Tapponnier, 1975; Besse et al., 1984; Patriat and Achache, 1984; Dewey et al., 1988; Chen et al., 1991, 1993; Le Pichon et al., 1992; Platt and England, 1993; Meyer et al., 1998; Yin and Harrison, 2000; Johnson, 2002).

Models proposed for the geodynamics of the Tibetan plateau fall into two main categories (Tapponnier et al., 2001): the first explanation involves a continental extrusion mechanics in which the crustal deformation results as a rigid indenter impinges into weak accreted crust and is characterized by localized shear between coherent lithospheric blocks with extrusion along large strike-slip faults (e.g. Molnar and Tapponnier, 1975; Tapponnier et al., 1982). The second explanation invokes continuous thickening with the indentation of India into Asia inducing broadly distributed thickening and shortening of the entire lithosphere in Tibet and areas to the north (e.g. England and Houseman, 1986; Dewey et al., 1989). Resolution between these two models requires knowledge of the kinematics of Tibet and adjoining parts of central Asia. The Global Positioning System (GPS) measurements can describe the recent velocity field in Tibet and its surroundings (e.g. Reigber et al., 2001; Wang et al., 2001) but it is not clear whether recent deformation rates can be extrapolated back over tens of millions of years to describe the evolution of Tibet and its surroundings (Dupont-Nivet et al., 2003). Paleomagnetism provides quantitative constraints on the kinematics of blocks since the formation age of the rocks, including the (paleo)latitudinal translation and vertical-axis rotation (e.g. Butler, 1992). However, available Cenozoic paleomagnetic data obtained from Tibet and central Asia are limited in number and it is difficult to construct detailed Cenozoic apparent

polar wander (APW) paths for individual blocks and to discuss the postcollisional kinematics in these areas. In addition, Cenozoic paleomagnetic data from central Asia (e.g. Chen et al., 1991, 1992, 2002a,b; Thomas et al., 1993; Cogné et al., 1995, 1999; Gilder et al., 1996, 2001, 2003; Halim et al., 1998; Rumelhart et al., 1999; Dupont-Nivet et al., 2002a,b, 2003) display paleomagnetic inclinations shallower than those computed from the master APW path of Eurasia by about 15–30°. Possible reasons for the shallow inclinations include non-dipole fields, a syndepositional inclination shallowing, initially shallow detrital remanent magnetization and postdepositional compaction-induced shallowing (Chen et al., 1993; Chauvin et al., 1996; Halim et al., 1998, Cogné et al., 1999; Gilder et al., 2001, 2003; Si and Van der Voo, 2001; Dupont-Nivet et al., 2002b; Tan et al., 2003).

Paleomagnetic results from intramontane basins of the Tian Shan orogen provide important constraints on the kinematics of this collisional mountain system. On the Chinese side of the Tian Shan, previous paleomagnetic study implies that little or no relative movement has taken place between the Turfan Basin and the adjoining Tarim Block following India-Asia collision although relative movement is likely to have occurred between Turfan and Junggar or Eurasia (Cogné et al., 1995).

To derive a better understanding of postcollisional vertical-axis rotation patterns on the Chinese side of the Tian Shan orogen, we have sampled Miocene red sedimentary rocks from the Turfan intramontane basin of the East Tian Shan, Northwest China for paleomagnetic study. Here we report and compare the results with data obtained from intramontane basins of the Tian Shan and evaluate kinematic patterns in the region.

2. Geological background and field sampling

The Tian Shan of central Asia is one of the world's great mountain ranges and extends east–west for at least 2500 km, with peaks exceeding 7400 m in height. The mountain chain was initiated in late Paleozoic times (Windley et al., 1990) but the remarkable modern elevation is related to the Cenozoic India-Asia collision and the postcollisional indentation of India into Asia (e.g. Molnar and Tapponnier, 1975; Hendrix et al., 1994).

The Turfan Basin is an elongated intramontane basin within the Tian Shan Range incorporating the second-lowest point of the Earth (−154 m) and located about 200 km east of Urumqi City of Xinjiang (Fig. 1b). Late Paleozoic, Mesozoic, and Tertiary deposits within the basin are mainly developed over the Huoyan Shan and the piedmont of the Bogda Shan (Fig. 2). The continental Tertiary system is divided into the Paleogene Shanshan Group, the Miocene Taoshuyuan Formation, and the Pliocene Putaogou Formation (TRGSX, 1978). The Shanshan Group is largely made up of reddish orange, brownish red, and pale-yellow sandstones and mudstones containing Ostracoda *Lipnocythere* (TRGSX, 1978). The Taoshuyuan Formation consists largely of reddish orange, brownish red, and white-gray sandstones, mudstones with conglomerates and/or coarse-grained sandstones in the lowermost part of the series. Typical fossils are Ostracoda *Candona subequalis*, *Cyclocypris cavernoas*, *Limnocythere iliensis* and Vertebrata such as *Amphechinus* sp., *Indricotherium grangeri*, *Parabrontops* sp., assigning a Miocene paleobiological age (BGMRX, 1993). The Putaogou Formation consists of brownish red, pale-red, and yellow sandstones intercalated with pale-yellow mudstones and conglomerates, with a layer of caliche conglomerates in the lowermost part, distinguishing the Pliocene sediments from the underlying Taoshuyuan Formation. This latter sequence contains the Pliocene Ostracoda *Eucypris concinna, Cypridopsis vidua, Zonocypris membranae* (BGMRX, 1993).

In the Turfan Basin the Paleogene sequences are generally in disconformable contact with both the underlying Upper Cretaceous strata and the overlying Miocene Taoshuyuan Formation, whilst the Miocene series is usually in conformable or disconformable contact with the overlying Pliocene Putaogou Formation. The Pliocene Putaogou Formation is conformable with the overlying conglomeratic Xiyu Formation (BGMRX, 1993), which is widely distributed along the margins of the Tian Shan Range and is assigned a Pliocene to Early Pleistocene age (e.g. Chen et al., 2002a). Folding is presumed to have occurred during the late Early Pleistocene (BGMRX, 1993) and produced a marked unconformity between the conglomeratic Xiyu Formation and the overlying

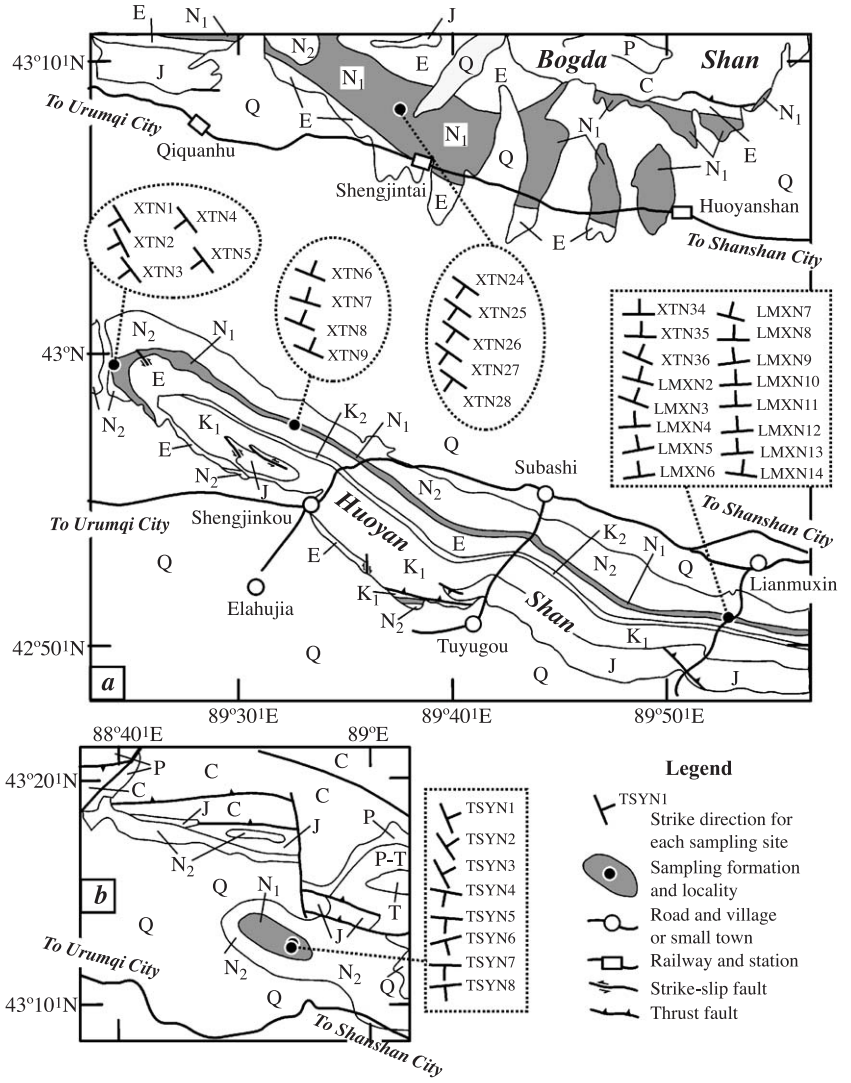


Fig. 2. Schematic geological maps of (a) the Huoyan Shan anticline and surroundings and (b) the Taoshuyuan sampling section, which is located at about 80 km west of the Shengjintai railway station, and surroundings showing post-Carboniferous stratigraphy and paleomagnetic sampling localities. Q, N_2, N_1, E, K, J, P, and C indicate Quaternary, Pliocene (Upper Neogene), Miocene (Lower Neogene), Paleogene, Cretaceous, Jurassic, Permian, and Carboniferous strata, respectively. Strike directions for each sampling site are shown in the insets.

middle Pleistocene sequence along the margins of the Tian Shan. In particular, it resulted in the formation of the Huoyan Shan anticline, as indicated by tilted pre-Quaternary strata overlain by flat-lying Pleistocene.

We collected 377 samples from five localities (40 sites) in Miocene red sedimentary rocks from the Turfan Basin (Table 1 and Fig. 2). In the Huoyan Shan anticline, 260 samples (27 sites) were collected from a locality West Huoyan Shan in the southwestern limb and from localities at Shengjinkou and Lianmuxin on the northeastern limb; 117 samples (13 sites) were collected from localities at Shengjintai and Taoshuyuan in the piedmont of the Bogda Shan (Fig. 2). Generally, each sampling site consists of about 10 independently drilled samples covering several meters of stratigraphic thickness cored with a portable drill and oriented using a magnetic compass.

Table 1
Summary of results of the high-temperature characteristic remanent magnetizations from the Turfan Miocene red sediments, East Tian Shan of Northwest China

Site ID	Strike/Dip	Range (°C)	R/N	n/n_0	D_g (°)	I_g (°)	D_s (°)	I_s (°)	k	α_{95}
West Huoyan Shan locality (42.98°N/89.40°E)										
xtn1	148.5/25	400–690	7/0	7/9	187.6	−4.9	183.1	−20.1	56.0	8.1
xtn2	152.5/33	400–670/690	0/9	9/11	44.0	3.1	40.0	34.2	31.4	9.3
xtn3	142.5/32	400–680/690	0/6	6/10	24.0	1.1	19.5	28.7	43.6	10.3
xtn4	142.5/32	400–680/690	0/5	5/17	352.8	2.5	347.6	17.7	96.9	7.8
xtn5	142.5/32	500–650/690	0/10	10/10	26.1	5.5	20.4	33.6	17.4	11.9
Averaging by sites			(1/4)	(5/5)	18.9	3.6			17.5	18.8
							13.2	28.0	18.1	18.5
McFadden's fold test (1990): N=5: in situ: ξ_2=0.836; tilt-corrected: ξ_2=0.115, critical value of the test statistic ξ=2.609 at the 95% confidence level indicative of indeterminate fold test.										
Shengjinkou locality (42.97°N/89.55°E)										
xtn6[a]	291.5/31	400–670/695	0/6	6/8	352.0	68.7	8.1	39.5	36.3	11.3
xtn7[a]	286.5/54	400–650/690	2/0	2/6	186.0	−72.7	193.2	−18.9	–	–
xtn8[a]	292.5/55	500–685/690	6/0	6/9	226.7	−69.5	211.1	−16.0	71.0	8.0
xtn9[a]	293.5/64	450–670/690	0/8	8/8	335.7	57.0	359.7	2.3	50.5	7.9
Averaging by sites			(2/2)	(4/4)	0.5	69.2			36.4	15.4
							13.1	19.5	16.7	23.2
McFadden's fold test (1990): N=4: in situ: ξ_2=0.463; tilt-corrected: ξ_2=2.777, critical value of the test statistic ξ=2.335 at the 95% confidence level indicative of negative fold test.										
Shengjintai locality (43.17°N/89.62°E)										
xtn24	125.5/27	450–640/690	5/0	5/6	183.2	−9.8	177.1	−32.0	32.0	13.7
xtn25[b]	125.5/27	500–655/685	4/0	4/7	185.4	−35.5	167.2	−56.8	20.4	20.8
xtn26	125.5/27	450–655/685	7/0	7/7	191.0	−23.9	181.3	−47.6	24.5	12.5
xtn27[b]	125.5/27	450–640/690	4/0	4/7	202.1	−9.0	199.2	−35.2	6.5	39.2
xtn28	125.5/27	530–640/690	6/0	6/7	190.6	−22.2	181.5	−45.9	40.6	10.6
Averaging by sites			(3/0)	(3/5)	188.2	−18.7			85.6	13.4
							179.8	−41.9	85.6	13.4
McFadden's fold test (1990): N=3: in situ: ξ_2=0.928; tilt-corrected: ξ_2=0.969, critical value of the test statistic ξ=2.076 at the 95% confidence level indicative of indeterminate fold test.										
Lianmuxin locality (42.85°N/89.88°E)										
xtn34	269.5/28	500–685/695	0/7	7/7	339.0	45.8	344.6	19.0	47.5	8.8
xtn35	272.5/28	450–670/695	0/3	3/5	6.2	49.6	5.1	21.6	130.0	10.9
xtn36	292.5/23	560–670/695	0/5	5/7	0.8	47.4	6.4	25.6	48.4	11.1
xtn37	271.5/29			0/9	No high-temperature component isolated					
lmxn1	234.5/35			0/8						
lmxn2[b]	284.5/38	300–590/610	0/5	5/8	353.7	47.2	0.3	10.8	24.9	16.7
lmxn3[b]	290.5/35	350–650/670	5/0	5/7	254.4	−67.1	224.2	−38.6	18.5	19.5
lmxn4	266.5/31	500–670/690	0/8	8/8	30.4	76.4	7.6	47.0	92.1	5.8
lmxn5	259.5/27	400–670/690	0/4	4/8	4.9	73.9	355.7	47.3	112.0	8.7
lmxn6	262.5/30			0/6	No high-temperature component isolated					
lmxn7	284.5/32	300–610/630	0/8	8/8	13.8	72.6	14.2	40.6	68.4	6.7
lmxn8	270.5/28	400–670/690	5/0	5/8	173.9	−62.1	176.8	−34.2	527.6	3.3
lmxn9	262.5/18	350–670/690	8/0	8/8	179.4	−69.0	176.4	−51.1	218.2	3.8
lmxn10	267.5/20	250–670/690	0/8	8/8	8.7	56.7	5.2	37.0	26.6	10.9
lmxn11	267.5/21	300–590/610	0/6	6/8	4.4	58.7	2.0	37.8	69.0	8.1
lmxn12[c]	265.5/12	350–690	0/6	6/8	81.1	23.5	76.1	22.1	21.6	14.7
lmxn13	265.5/12	350–690	0/6	6/7	37.3	56.0	28.2	46.4	30.9	12.2

(continued on next page)

Table 1 (*continued*)

Site ID	Strike/Dip	Range (°C)	*R*/*N*	n/n_0	D_g (°)	I_g (°)	D_s (°)	I_s (°)	*k*	α_{95}
lmxn14[c]	277.5/23	300–690	0/8	8/9	77.1	43.3	61.2	32.2	40.1	8.8
Averaging by sites			(2/9)	(11/18)	5.1	61.7			36.2	7.7
							3.4	37.6	33.3	8.0
McFadden's fold test (1990): *N*=11: in situ: ξ_2=1.923; tilt-corrected: ξ_2=2.862, critical value of the test statistic ξ=3.865 at the 95% confidence level indicative of indeterminate fold test.										
Taoshuyuan locality (43.21°N/88.92°E)										
tsyn1	337.5/22	350–690	0/7	7/8	337.3	42.2	356.1	38.6	131.6	5.3
tsyn2	326.5/31	300–670/690	0/4	4/7	347.8	62.6	21.0	42.4	142.0	7.7
tsyn3	332.5/24	300–670/690	0/8	8/8	347.2	55.4	12.9	43.9	124.2	5.0
tsyn4	100.5/68	450–670/690	0/5	5/7	6.0	−46.0	7.1	21.9	200.7	5.4
tsyn5	94.5/76			0/8	No high-temperature component isolated					
tsyn6	76.5/60			0/8						
tsyn7[c]	91.5/64	350–610/630	0/5	5/7	53.5	10.9	84.5	38.8	35.9	12.9
tsyn8	84.5/56	400–670/690	7/0	7/8	177.0	31.4	176.9	−24.6	89.1	6.4
Averaging by sites			(1/4)	(5/8)	351.5	19.3			2.6	59.8
							6.2	34.6	36.8	12.8
McFadden's fold test (1990): *N*=5: in situ: ξ_2=4.889; tilt-corrected: ξ_2=3.561, critical value of the test statistic ξ=3.573 at the 99% confidence level indicative of positive fold test.										
Overall mean by sites			(7/17)	(24/40)	7.2	36.8			5.3	14.2
							5.9	35.5	29.9	5.5
McFadden's fold test (1990): *N*=24: in situ: ξ_2=22.228; tilt-corrected: ξ_2=7.082, critical value of the test statistic ξ=8.007 at the 99% confidence level indicative of positive fold test.										

Abbreviations are: site ID, site identification; Strike/Dip, azimuth of right-hand strike and angle of dip of beds; *R*/*N*, samples show reversed and normal polarity; n/n_0, number of samples (sites in parentheses) joined the calculation/demagnetized; D_g, I_g (D_s, I_s), declination and inclination of direction in-situ (after tilt-correction); The overall mean by sites is calculated after the directions were transferred to the Lianmuxin locality of 89.88°E, 42.85°N, Turfan Basin.

[a] Sites were discarded because of a suspicion of remagnetization.

[b] Sites were rejected because of their large uncertainties (α_{95}>15°).

[c] Sites show significant divergent declination with respect to the other site-means.

Locations of the sites were determined by a handheld GPS receiver (Table 1).

3. Paleomagnetic study

The drilled samples were trimmed to cylindrical specimens of 2.2 cm length and analyzed at the paleomagnetic laboratory of the Institute of Geology and Geophysics, Chinese Academy of Sciences. Remanent magnetizations were measured by 2G Enterprises cryogenic magnetometer (2G-755R) and Characteristic Remanent Magnetizations (ChRMs) isolated following progressive demagnetization by MMTD60 thermal demagnetizer with residual magnetic field of less than 10 nT in the center of the oven. All the instruments are installed in a magnetic-free room, with maximum magnetic field less than 300 nT inside. In general, thermal demagnetization intervals are 100 °C at the lower temperatures reduced to intervals as small as 10 °C at the higher temperatures. Three hundred twenty specimens were subjected to progressively thermal demagnetization until the directional behavior became unstable due to acquisition of spurious magnetizations. Demagnetization results were evaluated using orthogonal diagrams (Zijderveld, 1967) and stereographic projections and magnetic components separated by principal component analysis (Kirschvink, 1980). Site and group mean directions were calculated using Fisher statistics (Fisher, 1953).

3.1. Thermomagnetic analysis of magnetic susceptibility

In order to explore the magnetic mineralogy of the sampled rocks, one or two pilot specimens per sam-

pling site, were subjected to thermomagnetic analysis of the magnetic susceptibility (MS) during a heating–cooling cycle from room temperature up to 700 °C in argon atmosphere, using a CS-3 apparatus coupled to the KLY-3 bridge.

Temperature dependence of the MS analysis shows that the heating and cooling curves of the MS of most specimens are roughly reversible, with a noticeable decrease of the MS around Curie temperatures of both magnetite and hematite (Fig. 3a–c). In addition, the heating and cooling curves of the MS of some specimens are irreversible, with a strong jump around 580 °C in the cooling curves (Fig. 3d–f), corresponding to formation of magnetite. Significant decrease of MS around the Curie temperatures of both magnetite and hematite is also observed. As the susceptibility of magnetite is about 1000 times greater than for most minerals (Collinson, 1983), this thermomagnetic behavior indicates that the studied rocks are dominated ferromagnetically by hematite; magnetite is present as a subordinate ferromagnet only.

3.2. Demagnetization results

3.2.1. West Huoyan Shan locality

Five sites (xtn1-5) were collected largely in muddy sandstones from roadcuts over a distance of about 1 km. Following removal of low unblocking temperature components (LTC) below a peak demagnetization temperature 350 °C, a concordant high-temperature component (HTC) was isolated from 39 out of 57 specimens demagnetized, between 500 and 650–690 °C (Fig. 4a, b and Table 1). In geographic coordinates, the LTC has an in-situ site-mean direction of D=329.9°, I=74.3°, α_{95}=16.7° (N=5, 31 specimens). The HTCs at four sites have normal polarity, whereas the remaining site has reversed polarity, with nearly antipodal direction (Figs. 4c, d

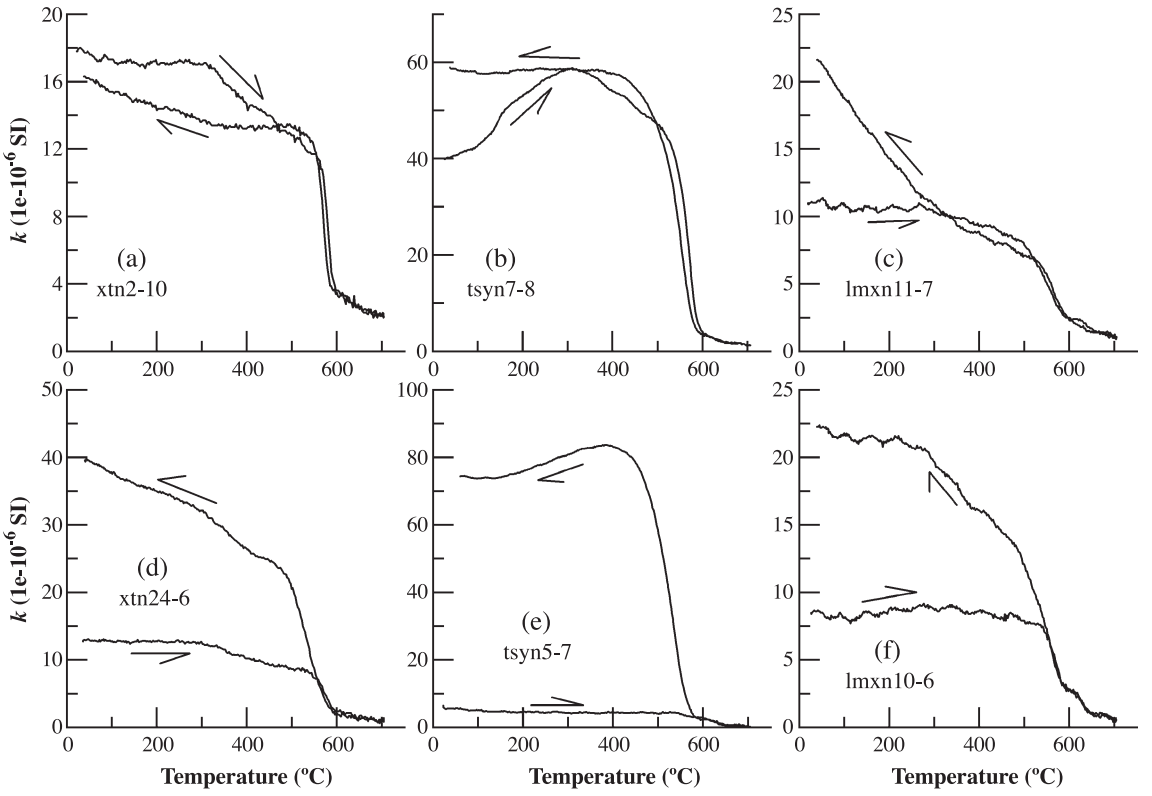


Fig. 3. Temperature dependence of magnetic susceptibility (MS) during a heating–cooling cycle from room temperature up to 700 °C in argon gas atmosphere for pilot Miocene specimens showing a ferromagnetic predominance of hematite with a subordinate presence of magnetite.

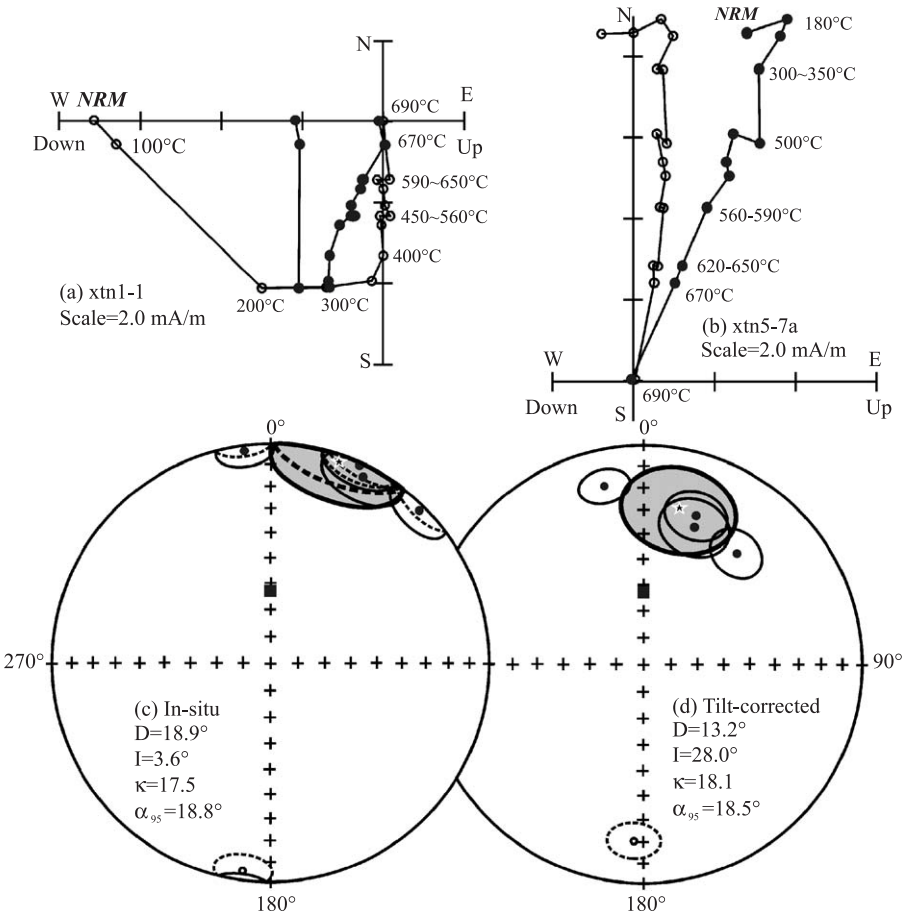


Fig. 4. Miocene red sediments from the West Huoyan Shan locality of Turfan, East Tian Shan of Northwest China. (a, b) Typical orthogonal vector plots. Directions are plotted in-situ, solid and open circles represent vector endpoints projected onto the horizontal and vertical planes, respectively. (c, d) Equal-area projections of the site-mean directions of the high-temperature component before and after bedding correction. Solid/open symbols represent downward/upward inclinations; stars indicate the overall-mean directions with 95% confidence limits; and squares indicate the present dipole field (PDF) direction.

and Table 1). The in-situ site-mean directions of the HTC are far from the Present Dipole Field (PDF) direction (Fig. 4c). A fold test (McFadden, 1990) is inconclusive at the 95% confidence level, since the dip of the bedding ranges from only 25–33° southwestwards (Table 1).

3.2.2. Shengjinkou locality

Following removal of a soft viscous overprint in the initial stage, progressive thermal demagnetization up to 690 °C at four sites (xtn6-9) revealed two magnetic components (Fig. 5a, b). The first-commonly removed below 350 °C has single polarity and clusters well around the PDF direction in geographic coordinates with an in-situ site-mean of D=0.5°, I=57.8°, α_{95}=15.2° (N=4, 31 specimens). Following removal of the LTC, a second magnetic component with dual polarities was isolated between 400/450 °C and 650 °C or higher temperature from most specimens. However, concordant HTC directions could only be isolated from two specimens at site xtn7. Four sites yielded an in-situ HTC mean direction of D=0.5°, I=69.2°, α_{95}=15.4° and a tilt-corrected direction of D=13.1°, I=19.5°, α_{95}=23.2° (Fig. 5c, d). There is noticeable decrease of precision parameter κ after the tilt correction (κ_g=36.4 versus κ_s=16.7) and a

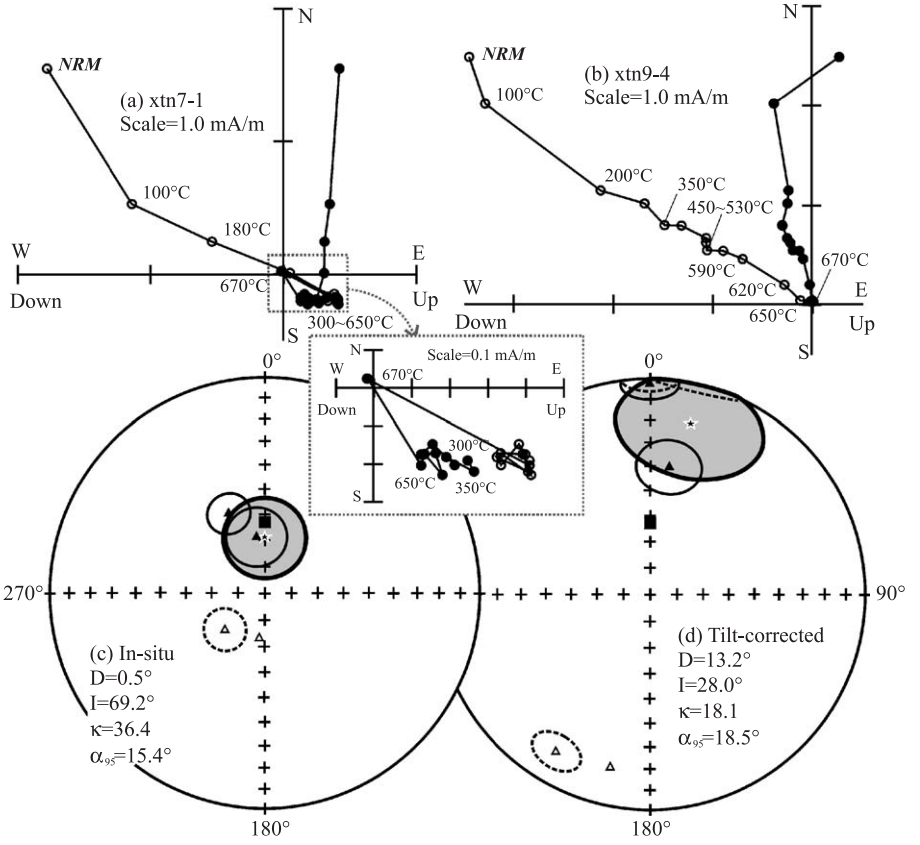


Fig. 5. Shengjinkou paleomagnetic locality. Symbols are as for Fig. 4.

negative fold test (Table 1). We therefore reject HTCs at these four sites from further analysis because complete remagnetization is suspected.

3.2.3. Shengjintai locality

Most specimens from five sites (xtn24–28) yiel ded two magnetic components after removal of an

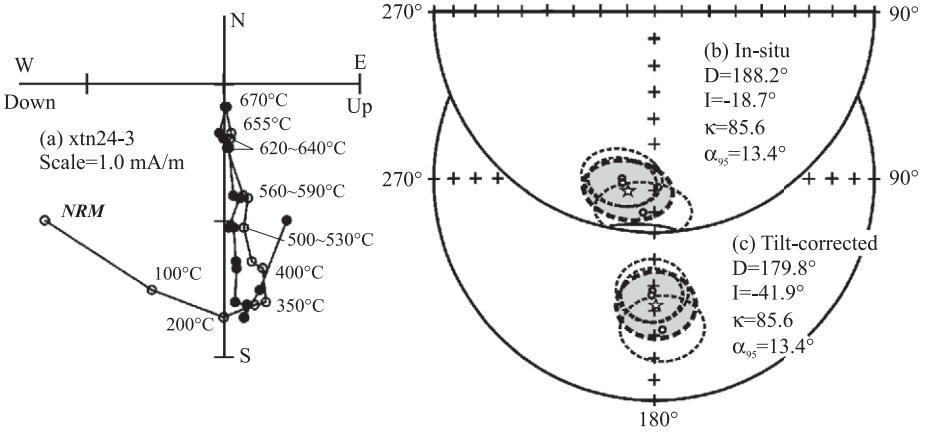


Fig. 6. Shengjintai paleomagnetic locality. Symbols are as for Fig. 4.

Table 2
Paleomagnetic poles from this study together with calculated flattening and rotation with respect to Eurasia, NCB, and Tarim at site 89.88°E, 42.85°N

Sampling position			Age	*N*	Pole position			$F\pm\Delta F$ (°)			$R\pm\Delta R$ (°)		
Locality	γ (°N)	ϕ (°E)			γ (°N)	ϕ (°E)	A_{95} (°)	Eurasia	NCB	Tarim	Eurasia	NCB	Tarim
W. Huoyan Shan	42.98	89.40	N_1	5	59.7	242.2	18.8	29.0±15.1	26.8±16.2	12.2±15.9	−7.9±15.6	−7.1±15.6	−14.7±15.6
								30.1±15.2		16.4±16.9	−6.3±15.6		−3.3±16.0
										13.1±16.0			0.4±15.6
Shengjintai	43.17	89.62	N_1	3	71.3	270.0	10.0	19.9±8.2	17.7±10.4	3.1±9.5	6.2±8.8	6.9±8.8	−0.6±8.8
								21.0±8.3		7.3±11.1	7.8±8.8		10.7±8.8
										4.0±9.6			14.5±8.8
Lianmuxin	42.85	89.88	N_1	11	68.6	260.5	7.3	22.4±6.1	20.2±8.4	5.6±7.8	2.4±6.3	3.2±6.3	−4.3±6.3
								23.5±6.2		9.8±9.6	4.1±6.3		7.1±6.3
										6.5±7.9			10.8±6.3
Taoshuyuan	43.21	88.92	N_1	5	65.7	253.2	11.4	24.7±9.3	22.5±11.0	7.9±10.5	−1.1±9.7	−0.3±9.7	−7.8±9.7
								25.8±9.4		12.2±11.9	0.5±9.7		3.6±9.7
										8.8±10.6			7.3±9.7
Mean by sites			N_1	24	66.8	255.1	5.1	23.8±4.4	21.6±7.3	7.0±6.5	0.1±4.3	0.8±4.3	−6.8±4.3
								24.9±4.6		11.3±6.7	1.6±4.3		4.6±4.3
										7.9±6.7			8.3±4.3

Block	Reference Poles	λ (°N)	φ (°E)	A_{95} (°)	Reference
Eurasia	N_1(10 Ma)	85.4	162.5	2.0	Besse and Courtillot, 2002
	N_1(20 Ma)	84.0	154.8	2.7	Besse and Courtillot, 2002
NCB	N_1(13–18 Ma)	76.4	178.0	6.0	Lin et al., 1985
	N_1(10–18 Ma)	86.6	236.1	4.9	Zhao and Coe, 1987
	N_1(14–25 Ma)	88.6	29.2	8.2	Zhao et al., 1990
	N_1(10 Ma)	83.4	192.8	6.2	Zheng et al., 1991
	N_1(mean)	84.8	188.6	7.6	This Study
Tarim	20 Ma	74.4	272.0	6.4	Rumelhart et al., 1999
	E_3–N_1 (20 Ma)	75.8	229.5	9.6	Chen et al., 2002b
	E_3–N_1	71.2	226.7	6.7	Chen et al., 1992

N, number of sites; A_{95}, radius of circle of the 95% confidence limit about paleomagnetic pole; *R* and *F* indicate vertical-axis rotation (+/−, counterclockwise/clockwise) and paleolatitudinal difference (+/−, northward/southward) with respect to Eurasian, NCB, and Tarim respectively; E_3 and N_1 indicate Oligocene and Miocene, respectively.

overprint in the initial stage of thermal cleaning (Fig. 6a). The LTCs removed, usually between 100 and 250–300 °C, cluster well around the PDF direction and have an in-situ site-mean of D=342.7°, I=69.0°, α_{95}=10.4° (N=5, 33 specimens). The HTCs were often removed between 400 and 670–685 °C,

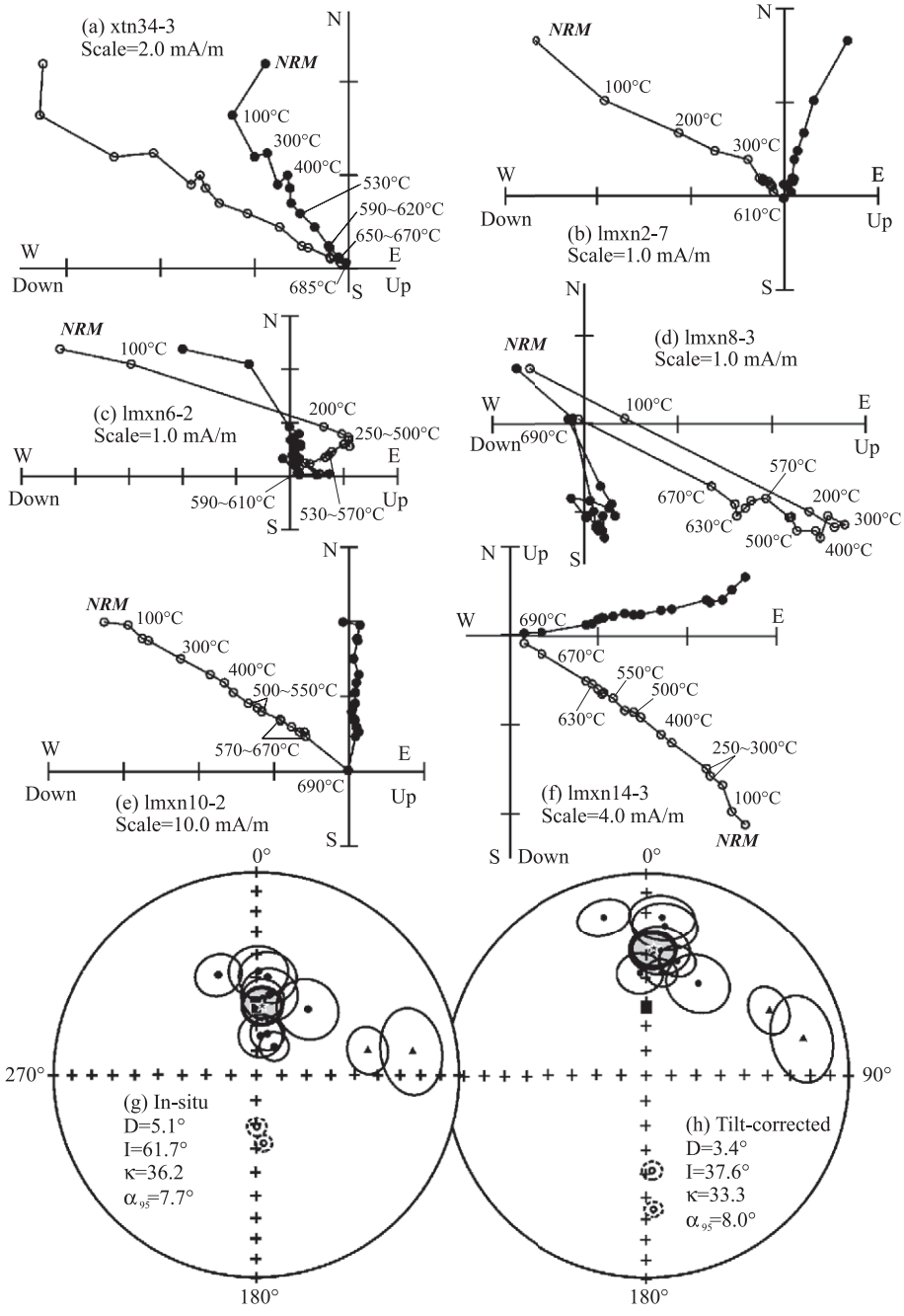


Fig. 7. Lianmuxin paleomagnetic locality. Triangles indicate sites showing divergent declination with respect to the other site-means. Symbols are as for Fig. 4.

and exhibit reversed polarity. However, the HTCs from specimens of sites xtn25 and 27 are rather scattered in declination and consequently provide site-mean observations with large uncertainties (α_{95}>15°, Table 1), which may be due to poor identification of components for these specimens. These two sites have therefore been discarded. The other three sites yielded concordant HTC directions (Fig. 6b, c).

3.2.4. Lianmuxin locality

18 sites were collected from this locality. Specimens from sites xtn34–37 and lmxn1–8 presented demagnetization behavior with a soft viscous overprint in the initial stage, the LTC removed usually below 250 and 350 °C, and the HTC removed between 400 °C or higher temperature and 590–690 °C (Fig. 7a–d). However, specimens from three sites xtn37, lmxn1 and 6 showed either an erratic

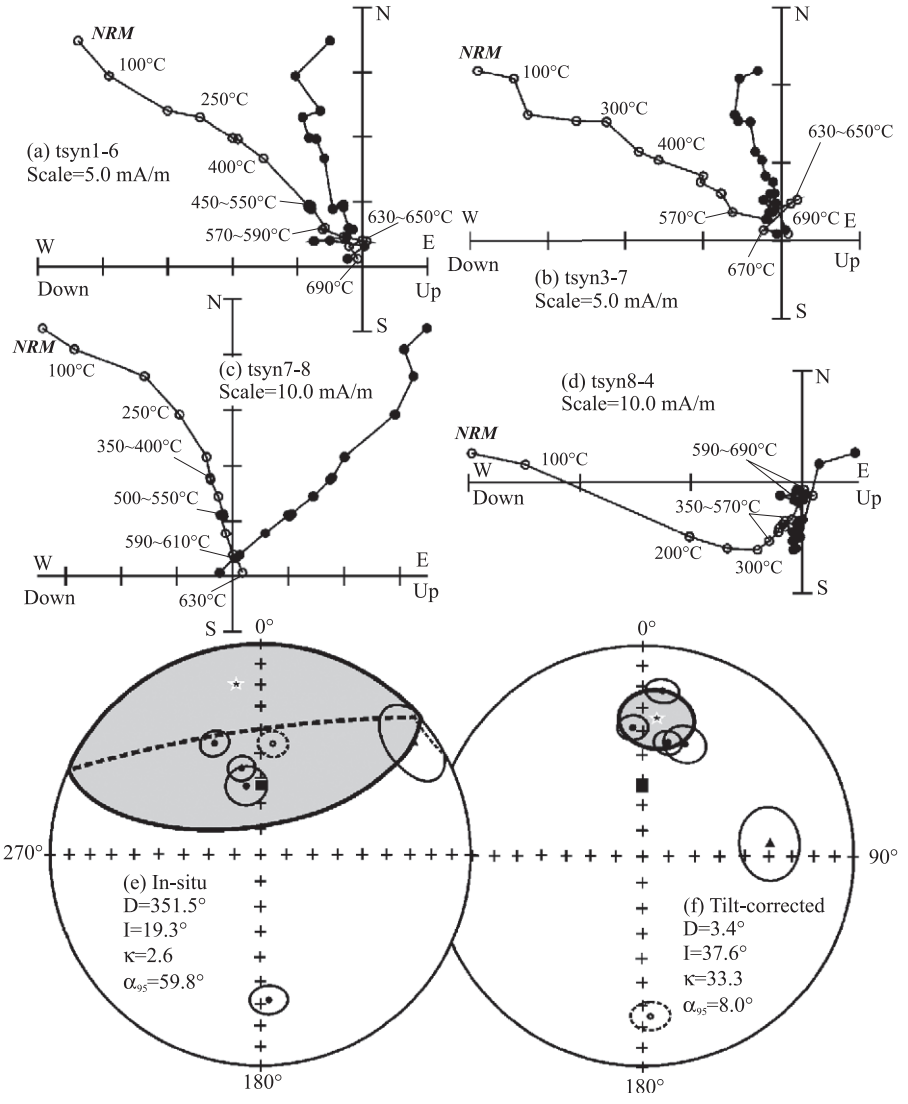


Fig. 8. Taoshuyuan paleomagnetic locality. Symbols are as for Figs. 4 and 7.

demagnetization behavior after removal of the LTC or poor separation of the HTC (Fig. 7c) and subsequently no stable high-temperature remanent magnetizations were isolated. Thermal demagnetization isolated a single component in most specimens from the remaining sites lmxn9–14 following removal of the LTC magnetization. In most cases, complete demagnetization occurred in peak temperatures between 670 °C and 690 °C (Fig. 7e, f) and we designate these ChRMs as the HTC (Table 1). The LTC directions are clustered around the PDF with an in-situ site-mean direction of D=351.1°, I=64.0°, α_{95}=5.3° (N=12, 79 specimens). The HTCs have dual-polarity with two sites revealing nearly antipodal directions (Fig. 7g, h and Table 1). Generally, the tilt-corrected site-mean directions point north to northeast with shallow to intermediate downward inclination and its antipode. However, the site-mean observations for sites lmxn12 and 14 have similar inclinations but divergent declinations (Fig. 7g, h and Table 2). We have no definite explanation for this except the possibility of local disturbance of the beds involved in these two sites and they were rejected from further analysis. Two further sites (lmxn2–3) provided site-mean observations with large uncertainties (α_{95}>15°, Table 1) and were discarded. The remaining 11 sites yielded concordant HTCs (Fig. 7g, h). The bedding dips uniformly and hence the fold test is not significant (Table 1).

3.2.5. Taoshuyuan locality

Eight sites (tsyn1–8) were collected from a kink fold in the piedmont of the Bogda Shan (Fig. 2b), and six of them yielded stable high-temperature remanent magnetizations (Table 1). During demagnetization, most specimens behave as specimens from sites xtn34–37 and lmxn1–8 of the Lianmuxin locality in that a LTC and HTC can be isolated (Fig. 8a–d). The former is generally removed by heating up to 350 °C. However, unblocking temperature of the LTC for some specimens from sites tsyn5–6 can extend to about 450 °C or higher temperature. We did not identify these remanent magnetizations as the HTC, because they are parallel to the LTC directions of the other specimens. Furthermore, most specimens yielded erratic demagnetization behavior after the LTC was completely removed and no stable HTC directions were isolated from these two sites (Table 1). The remaining six sites may yield adequately defined high-temperature remanent magnetizations. Nevertheless, the tilt-corrected site-mean direction of site tsyn7 shows a noticeable divergent declination with respect to the other site-means (Table 1 and Fig. 8b). Except for site tsyn8, all the specimens have normal polarity. Fisher statistics

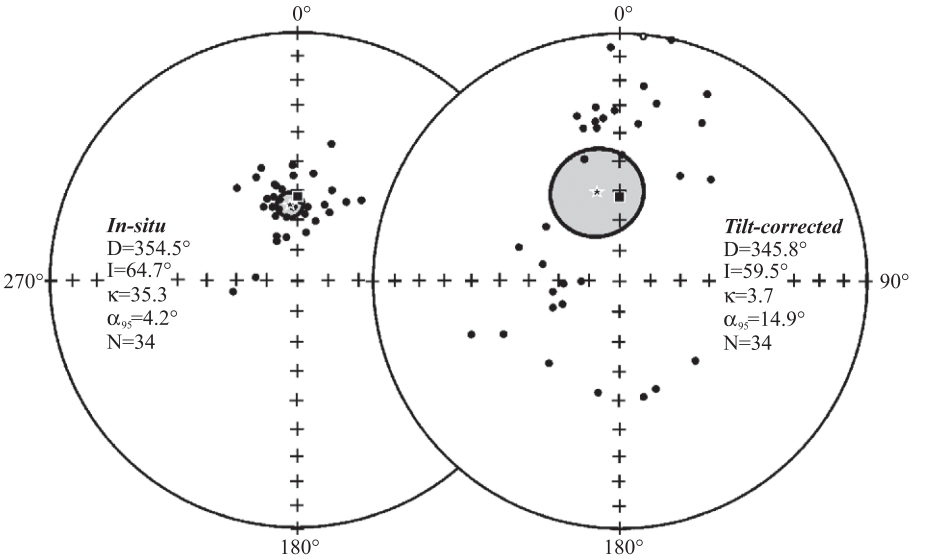


Fig. 9. Equal-area projection of the site-mean directions of the low-temperature component (a) before and (b) after bedding correction. Symbols are as for Fig. 4.

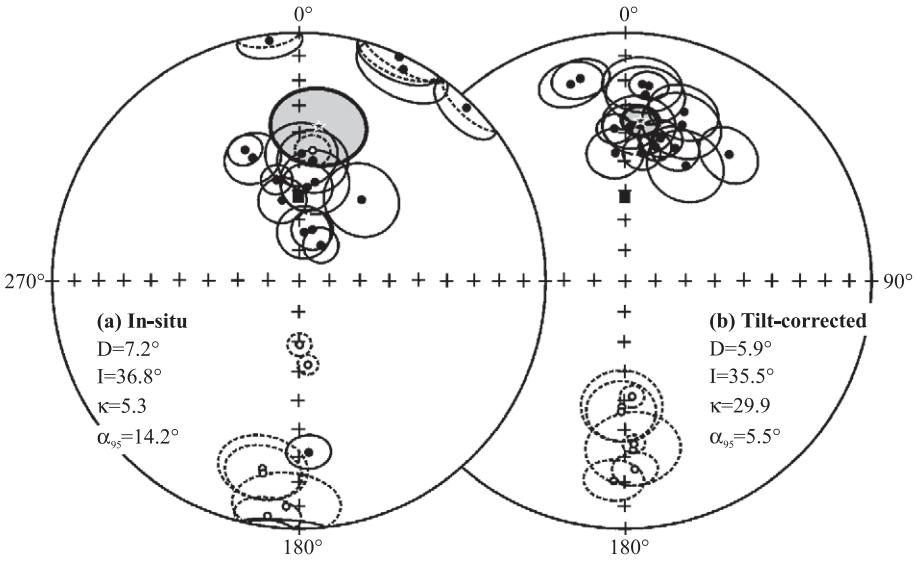


Fig. 10. Equal-area projection of the site-mean directions of the high-temperature component (a) before and (b) after bedding correction. Symbols are as for Fig. 4.

(1953) on five sites (excluding site tsyn7) provided a tilt-corrected mean of D=6.2°, I=34.6°, α_{95}=12.8° with statistically significant improvement in grouping (Figs. 8e, f and Table 1). When applying McFadden's fold test (1990), the site-means yields a positive fold test at the 99% confidence level (Table 1) indicating a pre-folding origin for these high-temperature remanent magnetizations.

In summary, most Miocene specimens exhibit two magnetic components following removal of a soft viscous overprint in initial stages of demagnetization. In the lower temperature stage a LTC was generally removed below 350–400 °C. The LTC has normal polarity and is evidently a recent overprint because it clusters around the PDF and a noticeable drop of precision parameter k occurs following tilt adjustment

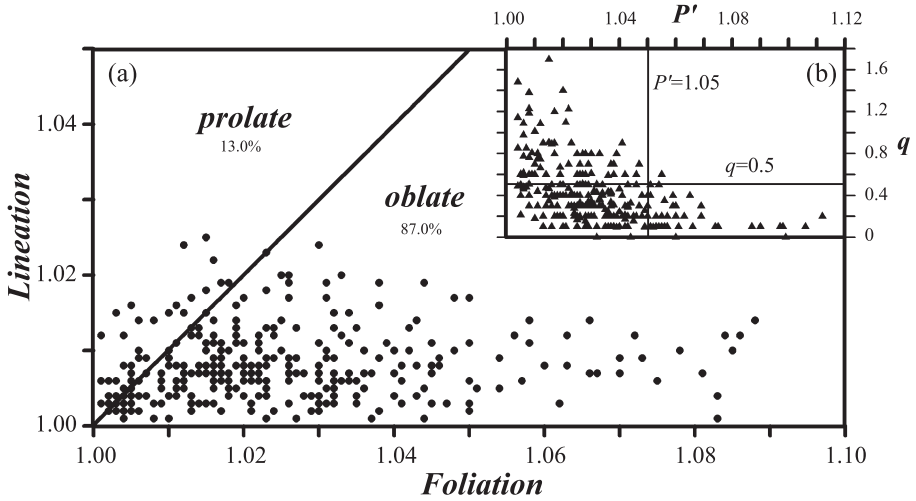


Fig. 11. Diagrams of AMS data for 324 Miocene specimens from the Turfan Basin showing (a) the MS foliation versus MS lineation, and (b) the corrected MS anisotropy (P') versus MS ellipsoid shape factor (q).

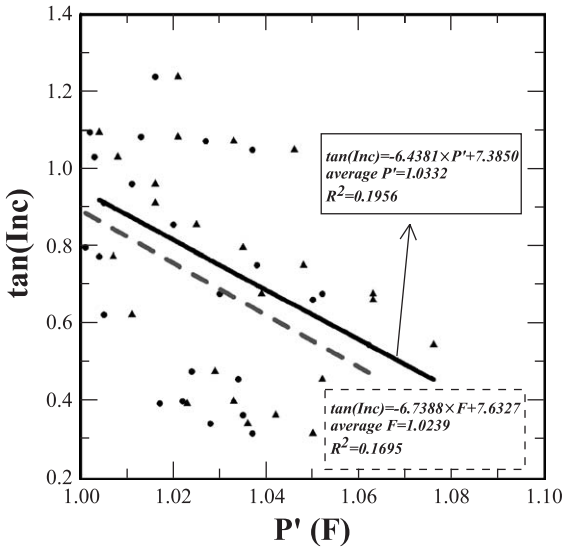


Fig. 12. Diagram of the 24 site-mean inclinations of the ChRM of the Turfan Miocene red sediments versus the corrected MS anisotropy (triangles) and MS foliation (circles) showing that the shallow inclinations are independent of the magnetic fabric.

(k_g/k_s=9.54, Fig. 9). With the exception of 16 sites that were discarded, 24 sites from four localities revealed concordant dual-polarity HTCs (Fig. 10 and Table 1). The magnetic mineralogy and demagnetization behavior suggest that this HTC resides in hematite. The fold test (McFadden, 1990) is positive at the 99% confidence level (Table 1) indicative of a pre-folding origin for this ChRM. The reversal test (McFadden and Lowes, 1981) is also positive and application of McFadden and McElhinny's (1990) classification of the reversal test shows a class C test (γ_0=7.8°) with a critical angle of γ_c=12.0° at the 95% confidence level. From the above analyses we interpret the ChRM to be a primary remanent magnetization acquired at, or close to, the time of formation of the rocks.

3.3. Anisotropy of magnetic susceptibility results

In order to check for a possible bias in paleomagnetic directions, due to ferromagnetic anisotropy we measured the anisotropy of magnetic susceptibility (AMS) using a KLY-3 bridge for 324 specimens, about eight per sampling site. The AMS data set displays a low anisotropy less than 1.05 for more than 80% of the specimens (Fig. 11b), averaging 1.0332 for specimens from the 24 sites that yield consistent ChRMs (Fig. 12). The mean value of the MS foliation is 1.0265 for all the measured specimens, while the MS lineation parameters have a relatively narrow range, and much smaller values than the MS foliation, averaging ~1.008 (Fig. 11a). For specimens from the 24 sites yielding ChRMs, the MS

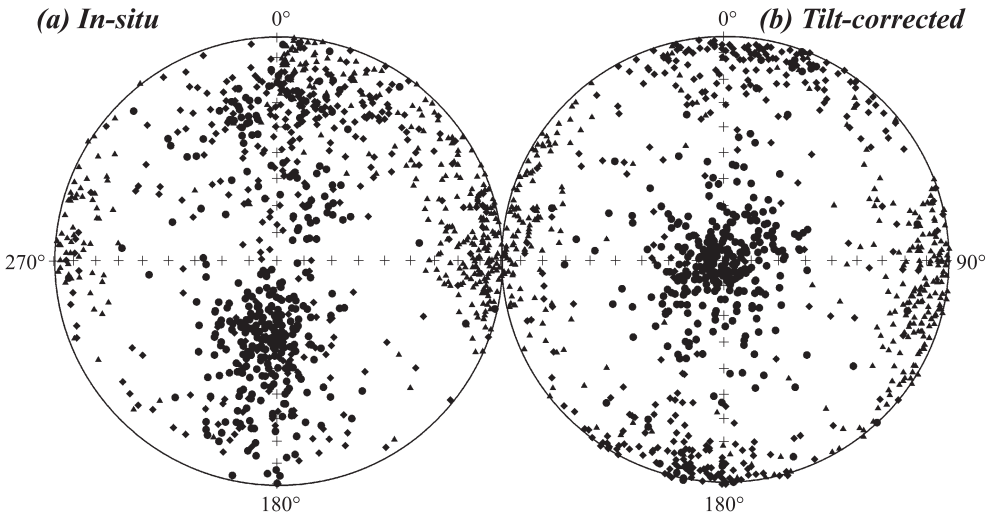


Fig. 13. AMS orientations for 324 Miocene specimens from the Turfan Basin. Stereographic projections show attitude of maximum (triangles), intermediate (diamonds), and minimum (circles) principal axes of the MS both (a) before and (b) after the tilt correction. All symbols are plotted on the lower hemisphere.

foliation has a mean value of 1.0239 (Fig. 12). The shape of the AMS ellipsoids is predominantly oblate (Fig. 11a). The tilt-corrected minimum MS directions are oriented roughly perpendicular to the bedding, while the maximum and intermediate MS directions are nearly horizontal and trend E–W and N–S, respectively (Fig. 13b). This magnetic fabric may predominantly be a depositional fabric (Hrouda, 1982 and references therein), because (1) the corrected anisotropy degrees P' are mostly lower than 1.05; (2) the q factors are generally lower than 0.5–0.7 (Fig. 11b); (3) the f (angle between the tilt-corrected minimum MS direction and bedding pole) factors are in the main lower than 15–30° (Fig. 13b); and (4) the grouping of the minimum MS directions is greatly improved after the tilt adjustment (Fig. 13). However, the clustered maximum susceptibility directions, which trend roughly parallel to the strike of the fold axis (102°N) defined by the bedding attitudes of the rocks, and the increased q and f factors in some specimens (Figs. 11b and 13b) suggest that the measured rocks are partially affected by weak deformation resulting from the N–S directed compression (Graham, 1966; Hrouda, 1982). From the above analyses we conclude that the measured rocks are unlikely to have experienced strong strain during compaction and tectonic stress.

4. Discussion

The high-temperature ChRM isolated from Miocene red beds in both limbs of the Huoyan Shan anticline and the piedmont of the Bogda Shan (Fig. 2) passes both fold (McFadden, 1990) and reversal tests (McFadden and Lowes, 1981; McFadden and McElhinny, 1990), and is consequently interpreted as a primary magnetization. For the entire data set, this ChRM has a tilt-corrected mean of D/I=5.9°/35.5° with α_{95}=5.5° (Fig. 10 and Table 1) equivalent to a paleomagnetic pole of 255.1°E, 66.8°N, with A_{95}=5.1° (Table 2).

4.1. Possible causes for inclination discrepancy

As shown in Table 2, the most striking feature common to our data is the inclinations ~20–30° shallower than those computed from both the 10 Ma and 20 Ma reference paleomagnetic poles for Eurasia (Besse and Courtillot, 2002). Anomalously shallow inclinations have repeatedly been observed in Cretaceous and Tertiary sedimentary rocks in central Asia Chen et al., 1992, 2002b; Thomas et al., 1993; Cogné et al., 1995, 1999; Gilder et al., 1996, 2001, 2003; Halim et al., 1998; Rumelhart et al., 1999, Dupont-Nivet et al., 2002a,b, 2003) and several hypotheses have been proposed for these discrepancies. First, relative convergent displacements between blocks in central Asia and Siberia could contribute to these discrepancies but are unlikely to have been large enough to explain ~20–30° anomalies since Miocene times (e.g. Dewey et al., 1989; Chen et al., 1993; Halim et al., 1998; Cogné et al., 1999). Recent paleomagnetic results on Cretaceous to Paleogene volcanic rocks from Southwest Tian Shan close to the China-Kyrgystan border (Bazhenov and Mikolaichuk, 2002; Gilder et al., 2003) have ruled out large northward tectonic transport, a non-dipole field, and inaccuracies in reference paleomagnetic poles for these anomalies. In addition, as Tan et al. (2003) discussed, the inaccurate age assignment to Asian sedimentary rocks can also not be blamed for all of the anomalous inclinations in central Asia, because at least some central Asian sedimentary rocks provided very typical Tertiary fossils and have detailed magnetostratigraphic constraints (e.g. Yin et al., 1998; Gilder et al., 2001; Chen et al., 2002a). Therefore, syndepositional inclination shallowing (Gilder et al., 2001, 2003) and rock-magnetic effects such as initially shallow detrital remanent magnetization or postdepositional compaction-induced shallowing (Dupont-Nivet et al., 2002b; Tan et al., 2003) might be more likely causes of the central Asian shallow inclinations.

The syndepositional inclination shallowing model was primarily devised to explain the shallow inclination in the Subei section of the Qaidam Basin. Shallowing was produced partly by imbrication during sedimentation under northward-directed transport conditions (Gilder et al., 2001). Gilder et al. (2003) further proposed that this hypothesis might also be valid when applied to the whole of central Asia. However, this hypothesis seems unlikely to explain the shallow inclinations in the Miocene rocks of Turfan, because the observed AMS orientations do not show imbrication (Fig. 13b). Moreover, it is not clear whether the southward-directed Lower Jurassic

paleocurrent in Turfan (Hendrix et al., 1992) had changed to a northward-directed transport direction by the Miocene.

We note that the mean MS foliation for the specimens yielding concordant ChRMs (1.0239) is smaller than that of Cretaceous red beds (1.035) in the Kuche Depression of Tarim (Tan et al., 2003) and that in Miocene sedimentary rocks (1.092) of the Maza Tagh Range of central Tarim (Dupont-Nivet et al., 2002b). However, all these Cretaceous to Miocene red beds exhibited ~20–30° of inclination shallowing. On the other hand, a plot of the 24 available site-mean inclinations of the ChRM of the Miocene specimens versus the mean MS anisotropies and foliations of each site reveals that the shallow inclinations are almost independent of the magnitude of the MS anisotropy and foliation (Fig. 12). In fact, since the bulk sample AMS may be affected by the AMS of diamagnetic calcite cement (Tan et al., 2003), isotropic red pigment, and a minor amount of magnetite (Bazhenov and Mikolaichuk, 2002), this independence cannot simply exclude the rock-magnetic effects (Dupont-Nivet et al., 2002b; Tan et al., 2003) as the main contributors to the shallow inclinations. However, postdepositional compaction-induced shallowing (e.g. Tan et al., 2003) is very unlikely to be the main contributor, because (1) the Miocene rocks in Turfan did not undergo strain during postdepositional compaction, and (2) the observed inclination discrepancies in central Asia do not show a latitude-dependence (e.g. Gilder et al., 2003). Our preferred interpretation is that the shallow inclinations in the Miocene rocks of Turfan is likely to be induced from an initially shallow detrital remanent magnetization (e.g. Dupont-Nivet et al., 2002b).

4.2. Vertical-axis rotation pattern of Turfan

In the Turfan Basin, we collected also 172 oriented drill-core samples from 17 sites from the Early Cretaceous Shengjinkou Formation in Shengjinkou and Tuyugou localities. Specimens from 13 sites yield a concordant high-temperature ChRM of D=8.8°, I=33.5°, k=50.3, α_{95}=5.9° after tilt correction. This ChRM possesses dual polarities and passes fold and reversal tests at the 95% confidence limit and is interpreted as a primary remanent magnetization (Huang et al., 2004). These directions combined together with previously reported data (Cogné et al., 1995) give a new Early Cretaceous paleomagnetic pole at 244.5°E, 65.3°N with A_{95}=5.0°. Comparison of the Turfan Miocene paleomagnetic pole with this new Early Cretaceous pole reveals non-significant or slight vertical-axis rotation of 4.8±4.3° during that period. On the other hand, as shown in Table 2, four localities distributed on both limbs of the Huoyan Shan anticline and the piedmont of the Bogda Shan (Fig. 2) show consistent non-significant vertical-axis rotations with respect to Eurasia, indicating no significant post-Miocene relative rotations between different localities of the Turfan Basin. This suggests that the Turfan Basin has not suffered strong structural deformation involving internal relative rotation since the Miocene. Indeed, with the exception of an inferred Turfan-Hami thrust fault at the southern margin of the Turfan-Hami intramontane basin, no large fault was known to crosscut this basin (Fig. 1b).

For the Turfan Basin as a whole, comparison of the Miocene paleomagnetic declination with those computed from the master APW path for Eurasia at 10 Ma and 20 Ma (Besse and Courtillot, 2002) reveals non-significant vertical-axis rotation of 0.1±4.3° and 1.6±4.3° between Turfan and Eurasia since the Miocene. Similarly, comparison of the Turfan Miocene direction with that expected from the mean Miocene pole for the North China block (NCB), which was calculated using four individual poles obtained from basalts ranging from ~10 to ~25 Ma (Lin et al., 1985; Zhao and Coe, 1987, Zhao et al., 1990; Zheng et al., 1991), also suggests no significant vertical-axis rotation of 0.8±4.3° between these two tectonic units since the Miocene (Table 2 and Fig. 14a). Moreover, non-significant or marginal significant vertical-axis rotation with respect to Eurasia is also observed from Paleogene rocks in the Issyk Kul and Naryn basins of West Tian Shan (Thomas et al., 1993). In contrast, paleomagnetic declinations from Lower Cretaceous to Miocene rocks suggest that blocks surrounding and including the Ferghana Basin have undergone a 20–30° counterclockwise rotation with respect to Eurasia (Bazhenov, 1993; Thomas et al., 1993). Taking the above analyses together, we argue that indistinguishable vertical-axis rotation with respect to Eurasia appears to be a common feature for the Tian Shan intramontane basins to the east of the Talas-Ferghana right-lateral strike-slip fault (Fig. 14b), though defor-

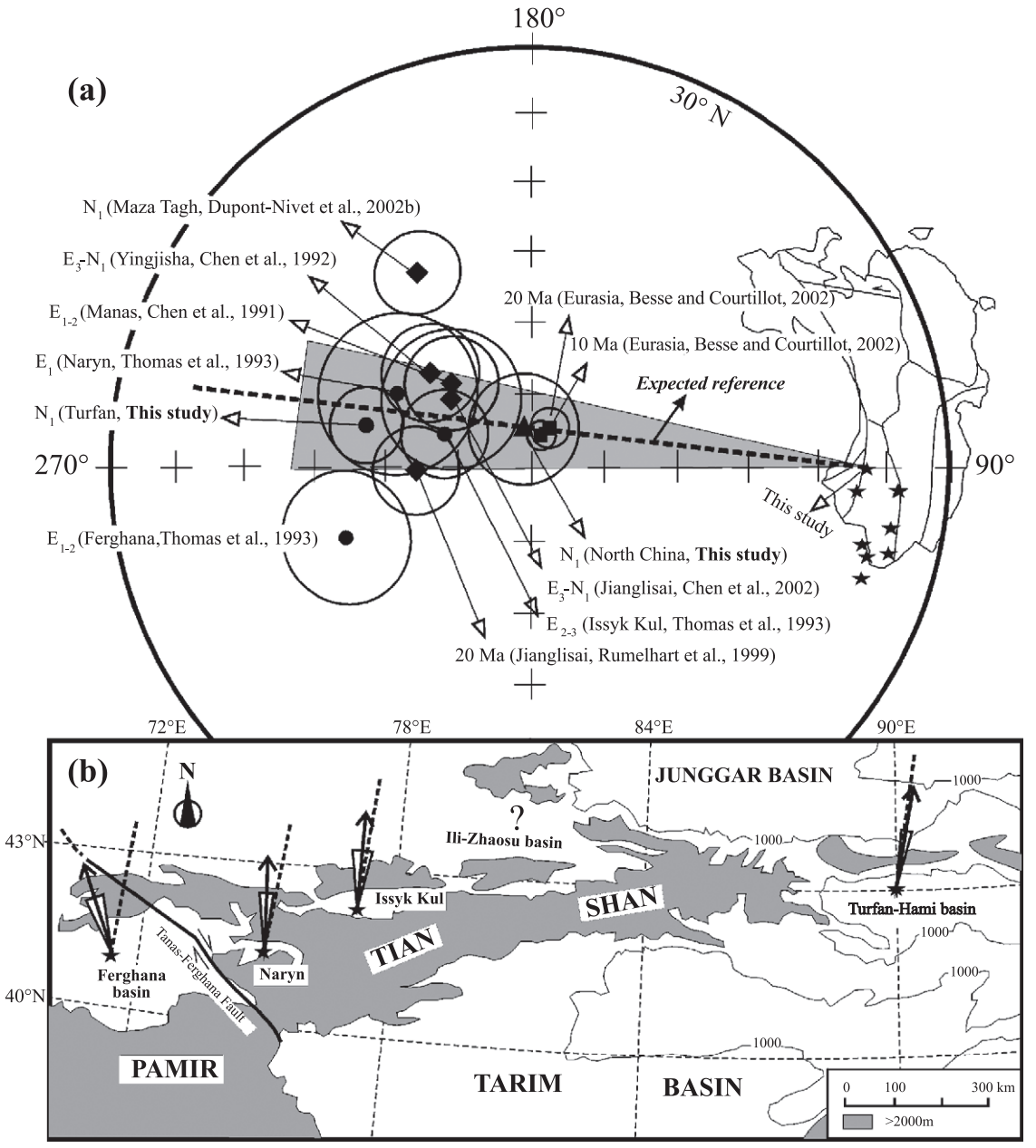


Fig. 14. (a) Equal-area projection of Miocene paleomagnetic references for Eurasia (squares) and North China (triangle) and available Tertiary paleomagnetic poles from the intramontane basins of the Tian Shan (circles), Tarim and Junggar (diamonds) showing little or no vertical-axis rotation between Turfan and the surroundings. Paleomagnetic poles are plotted onto the Northern Hemisphere with 95% confidence limits. (b) Map of the Tian Shan Range showing different vertical-axis rotation patterns in the intramontane basins on different sides of the Talas-Ferghana fault; contours every 1000 m. The arrows show paleomagnetic rotations with respect to Eurasia with uncertainties in triangle-shape designs. Simplified from Chen et al., 1991. Stars indicate the corresponding sampling locations of the available paleomagnetic poles from central Asia.

mation for the entire Tian Shan started in the Paleogene, and became more rapid in the Neogene and has continued to the present day (Thomas et al., 1993). However, this observation requires confirmation from further paleomagnetic study of Tertiary rocks from the Ili-Zhaosu intramontane basin of West Tian Shan (Figs. 1b and 14b).

Recently, two Miocene paleomagnetic studies have been carried out at the southeastern end of the Maza Tagh Range of central Tarim (Dupont-Nivet et al., 2002b), and in Jianglisai village near the southeastern border of Tarim (Chen et al., 2002b). The former study indicates a local vertical-axis rotation that cannot be used as a reference for the entire Tarim Basin. The latter study obtained a concordant paleomagnetic direction from seven Oligo-Miocene red bed sites, to which Rumelhart et al. (1999) assigned a mean age of 20 Ma. The available paleomagnetic results from Jianglisai at the southeastern border of the Tarim (Rumelhart et al., 1999; Chen et al., 2002b) indicate only slight vertical-axis rotations of $6.8\pm4.3^\circ$ and $4.6\pm4.3^\circ$ between Turfan and Tarim since the Miocene (Table 2 and Fig. 14a). Moreover, a Cretaceous pole from the Yingjisha Section near the southwestern border of Tarim (Chen et al., 1992), which was recently assigned to the Oligo-Miocene by the original authors (Chen et al., 2002b), indicates also a small vertical-axis clockwise rotation of $8.3\pm4.3^\circ$ between Turfan and Tarim (Table 2 and Fig. 14a). Taking the above analyses together, we argue that the Turfan intramontane basin is unlikely to have undergone significant vertical-axis rotation with respect to the Tarim Basin since the Miocene.

5. Conclusions

The present paleomagnetic study in the Turfan Basin shows an inclination $\sim20-30^\circ$ shallower than those derived from reference paleomagnetic poles for Eurasia at 10 and 20 Ma (Besse and Courtillot, 2002). The low MS anisotropy degrees, AMS shape and orientations suggest that the shallow inclination is likely induced from an initially shallow detrital remanent magnetization (e.g. Dupont-Nivet et al., 2002b).

Comparison of the Turfan Miocene paleomagnetic declination with those expected for Eurasia, NCB, and Tarim suggests that the Turfan Basin has not experienced significant vertical-axis rotation since the Miocene. Together with paleomagnetic declinations on both sides of the Talas-Ferghana fault, the Tian Shan to the east of the Talas-Ferghana fault appears to have remained relatively stationary with respect to adjacent blocks since the Miocene.

Acknowledgements

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